



ESSENCE OF TESTING

essenceoftesting.com

Anand Bagmar

Engineering Leader | AI-Enabled Engineering

Architecture, Delivery & Quality · Engineering Strategist · 25+ Years

Open to: FTE · Fractional · Consulting & Advisory

linkedin.com/in/anandbagmar · @BagmarAnand

Strategic where it matters. Practical where it counts.



Engineering Executive · Strategist · AI-Enabled Expert

Anand Bagmar is a software engineering executive, keynote speaker, and AI-enabled engineering practitioner with over 25 years of experience leading engineering strategy, architecture and transformation across complex technology environments - with deep expertise in quality engineering and test automation as part of that story.

His work spans engineering assessments, architecture and roadmap definition, AI-enabled delivery, and hands-on implementation across telecom, e-commerce, gaming, SaaS, enterprise platforms, and banking/financial services. A Selenium and Appium contributor and creator of open-source tools used globally — and a regular conference speaker on engineering effectiveness, AI-driven engineering, and software quality.

DOMAINS: Telecom · E-Commerce · Gaming · SaaS · Banking & Financial Services · Enterprise Platforms

25+

Years' Experience

15+

Global Clients

100s

Professionals Trained

3

Open-Source Tools



Six pillars of measurable impact

1

Engineering Leadership & Strategy

Engineering and quality operating models, governance frameworks, and risk-based practices that align architecture, delivery and quality with business outcomes.

2

AI-Enabled Engineering & Quality

AI-driven engineering and scalable automation for web, mobile, API, desktop, and visual testing — improving reliability across CI/CD pipelines.

3

Transformation & Advisory

Quality assessments, roadmaps, and contextual implementation — equally effective in short engagements and full programmes.

4

Training & Enablement

Workshops, bootcamps, and enterprise programmes — training hundreds of professionals on modern automation and AI adoption.

5

Framework & Tool Innovation

Creator of Teswiz, WAAT, and TTA — open-source tools adopted and trusted by the global testing community.

6

Cross-Functional Improvement

Works across product, engineering, and leadership teams to reduce friction and build scalable, sustainable QA capability.



Trusted by global leaders across industries

AmberPoint

AmDocs

Applitools

Borland

ChargedUp.green

Entain

Microsoft

Myntra

SnapDeal

Specmatic

Srijan

ThoughtWorks

Tieto

Vuclip

WebMD

Xnsio

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Building tools the global AI-engineering community relies on



Teswiz

End-to-End Automation Framework

Open-source, scalable engineering automation framework for web, mobile (Android & iOS), API, and desktop - including MCP support for AI agent workflows. Built for real-world CI/CD integration and used globally.

✓ Web & Mobile (Android / iOS)

✓ API & Desktop Testing

✓ CI/CD Native Integration

✓ Globally Adopted



WAAT

Web Analytics Automation Testing

Dedicated framework for automated testing of web analytics implementations — ensuring accurate data collection and event tracking at scale.

✓ Analytics Validation

✓ Event Tracking Automation

✓ Automated Assertions

✓ Scalable Architecture



TTA

Test Trend Analyzer

Intelligent analysis of test execution trends — surfacing patterns in stability, flakiness, and coverage to drive data-informed quality decisions.

✓ Trend & Pattern Analysis

✓ Flakiness Detection

✓ Coverage Insights

✓ Data-Driven Decisions



How Essence of Testing creates value

01 Strategic Consulting

Practical, fit-for-purpose solutions leveraging AI and industry best practices — tailored to your product, team, and business goals.

02 AI-Powered Solutions

Adopt innovative AI tools to streamline testing, accelerate automation, reduce flakiness, and improve overall reliability.

03 Team Upskilling

Transform manual testers to automation-ready contributors. Deliver better ROI from your automation investment through structured programmes.

04 QA Community Engagement

Stay ahead with the latest practices, tools, and thought leadership — connecting with the global quality engineering community.



Practical skills for real-world quality challenges

1 Quality Strategy in the AI-Accelerated World

Workshop for managers: building product confidence when delivery is faster, more distributed, and AI-supported.

2 Boosting Automation with AI

Apply AI practically to make automation smarter, faster, deterministic and resilient.

3 Future-Ready Automation Frameworks

Design scalable, maintainable frameworks that grow with your product and teams.

4 Quality Metrics in the Age of AI

Track what truly matters — modern metrics for intelligent, data-driven quality decisions.

5 Test Automation Coverage & ROI

Measure what your automation truly covers and the business value it delivers.

6 Automation Framework 101

Learn the fundamentals of building robust automation for web and mobile applications.

7 Visual AI for Test Coverage

Expand and accelerate UI test coverage for web and mobile using Visual AI tools.

✦ **Custom programmes:** Workshops and training programmes can be tailored to your team's specific requirements, existing challenges, or technology stack.



Practical skills for engineering and quality challenges

1

Architecture for Scale: Designing Test Automation That Grows 10x

Design test architecture that scales as your product, team, and integrations grow – across platforms (API, web, Android, iOS)

2

AI-Era Engineering Effectiveness: Measuring What Actually Matters

Move beyond vanity metrics to the few signals that predict delivery confidence and engineering health in the AI era.

3

AI Agents & MCP for Engineering Teams

Practical grounding in agentic workflows and MCP servers - what to trust, verify, and keep reliable in production.

4

Contract Testing with Specmatic

Catch API drift before it reaches a shared environment - contract testing fundamentals and adoption patterns.

5

CI/CD Pipeline Generation

Generate your CI configuration from the same domain model that drives your tests, so it never silently drifts out of sync.

6

Multi-Platform Test Architecture

Write a test once against business intent and run it unchanged over API, web, and mobile (Android/iOS).

7

Automation Engineer to Architect

A career-track workshop for senior ICs ready to move from writing automation to designing systems that make it scale.

✦ **Custom programmes:** Workshops and training programmes can be tailored to your team's specific requirements, existing challenges, or technology stack.



Influencing engineering and quality worldwide

25+

Years of QA Excellence

100s

Professionals Trained

15+

Global Organisations

3

Open-Source Tools



Keynote Speaker

Regular speaker at major global conferences on engineering effectiveness, AI-enabled engineering, and quality.



Published Author

Thought leadership through published articles and community contributions on engineering, testing strategy, and AI.



Open-Source Creator

Selenium and Appium contributor. Creator of Teswiz, WAAT, and TTA — tools trusted by the global engineering and QA community.



Keynote: Power Your Automation using AI



1,080 Combinations, 15 Real Ones

DOMAIN**Gaming****TYPE****Architecture & Test Automation****ONBOARDING****~1 Hour****SCALE****6 Integrations · 88 Scenarios****Objective**

- Eliminate the recurring, multi-day cost of onboarding each supplier integration
- Replace a combinatorial explosion with a model matching real business risk
- Make it safe to add the next integration without disturbing anything shipped

Approach

- The matrix is a tree, not a product — 1,080 combinations collapse to 15 real ones
- Draw a hard boundary between business intent and technical wire format
- Generate the CI pipeline from the same domain model that drives the tests

Outcomes

- Integration onboarding: days of manual work, every cycle → ~1 hour, once, automated
- Zero regression risk — a new integration ships in one commit, touching no files
- Architecture built for a 100x roadmap — 108,000+ combinations across API, web, mobile



Gaming Supplier Integration Automation — Complexity at Scale

From a difficult integration matrix to a massive **multi-channel**, multi-market automation challenge

Original Case Study

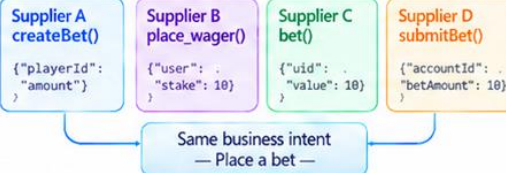
A focused but complex integration matrix



$$6 \text{ integrations} \times 5 \text{ brands} \times 3 \text{ markets} \times 3 \text{ environments} \times 4 \text{ games} = 1,080 \text{ combinations}$$

Same business intent, different technical dialects

Different supplier APIs.
Same underlying business meaning.

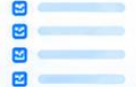


Model the matrix correctly

1,080 nominal combinations collapsed to 15 real ones



88 scenarios



5-way parallel execution

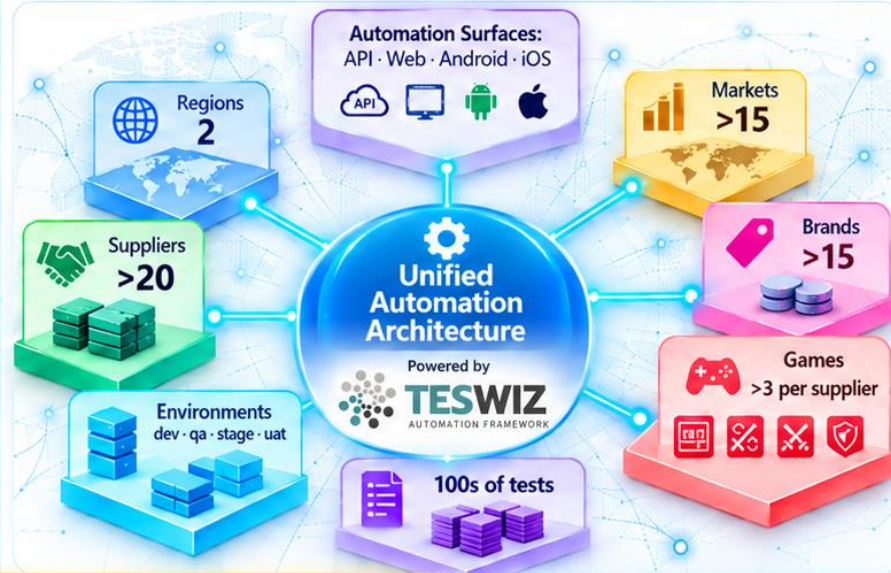


New integration onboarding

days → ~1 hour

Grand Goal

A massive multi-channel, multi-market automation challenge



Minimum execution contexts per test

> 108,000



2 regions × 15 markets × 15 brands × 20 suppliers × 3 games × 4 environments



A API, Web, Android and iOS are counted separately.
>432,000 contexts per test



100s of tests = tens of millions of potential executions



Any valid combination. Any environment. One scalable automation strategy.



Upskilling 50 SDETs with AI-Powered Test Automation

DOMAIN

Gaming

TYPE

Classroom Training

DURATION

5 Weeks

SCALE

50 SDETs



Objective

- Upskill 50 SDETs to leverage AI tools and modern automation technologies
- Enable meaningful contribution to existing Test Automation Platform
- Accelerate the path to automation-ready contributors via practical training

Approach

- Rigorous classroom-based programme with customised curriculum
- AI-assisted automation, daily evaluations, and final projects
- Technical mentoring and feedback integrated throughout

Outcomes

- Transitioned testers from manual to automation-ready roles
- Confidence in writing, reviewing, and scaling test scripts with AI
- Multiple participants demonstrated leadership potential in automation



Reliance Jio: Transforming Quality Engineering at Scale

DOMAIN

Telecom · Retail · eComm · Media

TYPE

Assessment · Strategy · Impl.

DURATION

4.5 Years

SCALE

20+ Teams | 75+ SDETs

Assessment & Strategy

Reviewed existing practices, identified bottlenecks, and created an actionable roadmap aligned with business goals.

Automation Frameworks

Built reusable frameworks for web, mobile (Android/iOS), desktop, and APIs — integrated with Azure DevOps quality gates.

AI Visual Testing

Implemented AI-driven visual testing where applicable, improving the reliability of visual validations at scale.

QECC Establishment

Standardized tools, frameworks, and processes. Sourced, recruited, and trained 75+ SDETs with competency mapping and project integration.



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Let's connect.

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Website: essenceoftesting.com

GitHub: [@anandbagmar](https://github.com/anandbagmar)

Twitter: [@BagmarAnand](https://twitter.com/BagmarAnand)

LinkedIn: linkedin.com/in/anandbagmar